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VYOMA DESIGN-A-THON

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Problem Statement 4

Smart Modular Workstation for Disaster Relief

Bringing Order to Chaos, One Module at a Time.



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Vision

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Natural disasters often leave communities isolated, without power, medical resources, or communication capabilities.

We propose the Smart Modular Workstation—a versatile, air-droppable platform designed to bridge that critical gap. By integrating four specialized modules into one compact unit, our solution delivers immediate energy, medical support, water purification, and survival tools.

This modular architecture allows rapid configuration to meet diverse mission requirements, ensuring first responders and affected populations receive essential infrastructure the moment it lands.



- Endurance: 36–49 hours
- Drop velocity: ~18 m/s post-parachute
- Mass: ~75 kg total
- Operating range: -10 °C to 40 °C



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Mission Use Case

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After a major natural disaster disables communication infrastructure in a remote area, a Smart Modular Workstation is air-dropped from a helicopter at 1000 meters. Guided by onboard navigation, it lands precisely with a parachute slowing descent to 18 m/s and airbags cushioning impact for safe deployment.

Upon landing, the workstation automatically activates, deploying solar panels for power and establishing a satellite link for GPS tracking and emergency ham radio communication.

Relief teams use it as a mobile command center, coordinating drone-based aerial surveys via an IoT dashboard. Medical supplies and a water purification system in Module 2 provide immediate aid. Module 3's thermal imaging helps locate hidden or trapped individuals.

Engineered for rapid response, the workstation enhances rescue efficiency, situational awareness, and helps restore vital services in the critical early hours post-disaster.



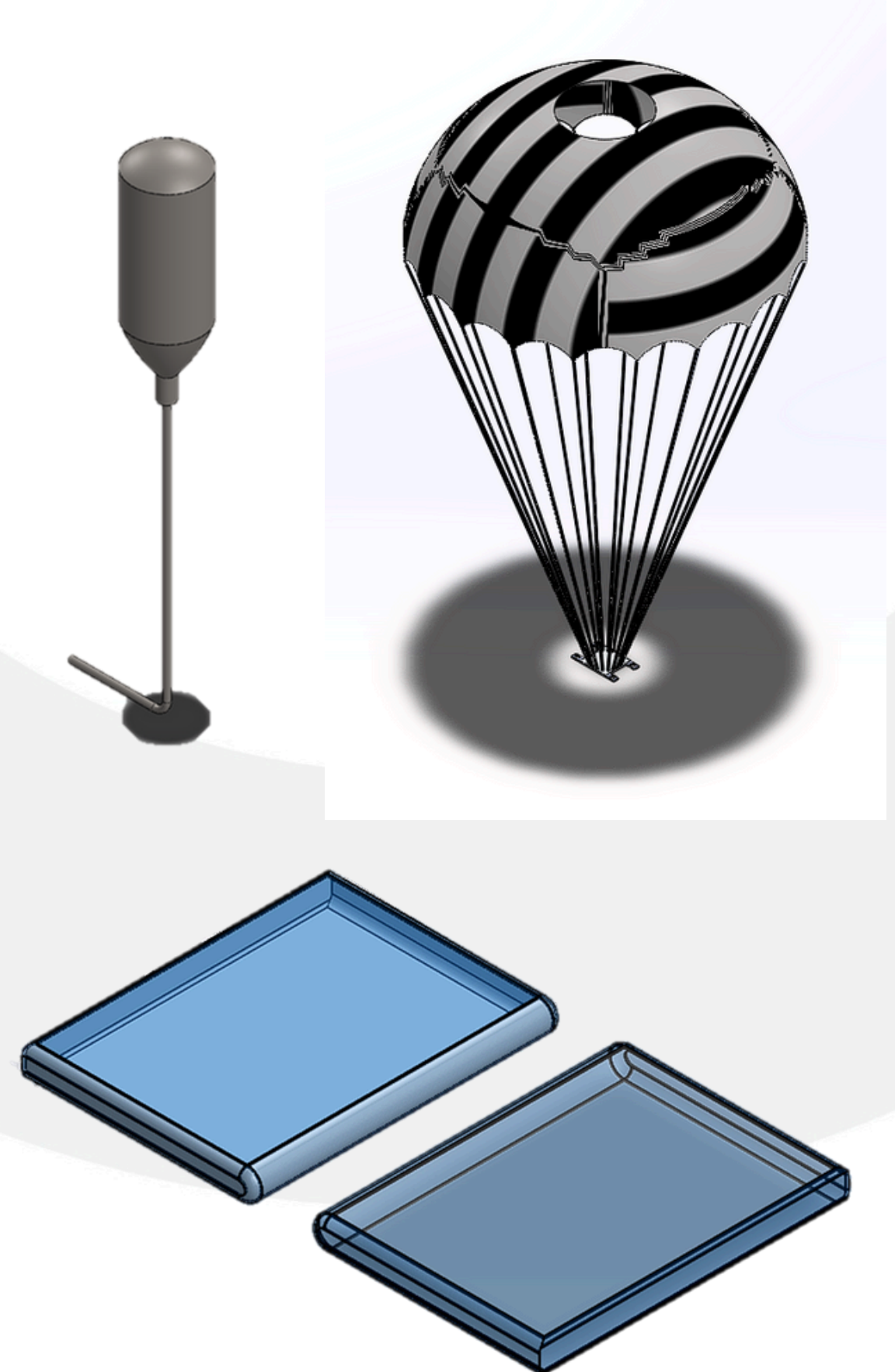


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Module 1 – Deployment Mechanism *Go, change the world®*

This module ensures safe delivery and operational readiness:

- **Descent System:** Air-drop enabled via helicopter or large drone; descent guided by Raspberry Pi with integrated GPS and IMU sensors.
- **Controlled Landing:** The parachute deploys at a particular altitude; main parachute (C_e 1.75, A 1.8237m²) reduces descent velocity to ~6.2m/s.
- **Impact Mitigation:** CO₂-inflated automotive airbags and crushable foam cushion landing forces up to 30kN at 20m/s, protecting internal payload.
- **Payload Security:** Internal modules are fixed within the chassis to withstand high-impact forces; system can be disassembled and retrieved post-mission.
- **Solar Panel Array:** Foldable, lightweight panels auto-deploy using spring-loaded hinges post-landing, providing sustainable power for immediate operation.





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Module 2 – Power & Communication

This core module provides electrical power and long-range communication capabilities. Positioned at the base for optimal weight distribution, it comprises:

- **Battery System:** Includes integrated charging/discharging circuitry.

A LithiumPolymer battery has been chosen for its high energy density and flexible form factor.

- **IoT-Enabled Raspberry Pi:** Facilitates real-time monitoring of power status, GPS location, and occupancy data through a dashboard interface. 2 The Raspberry Pi 5 weighs 50 grams, and along with casing 74 grams.

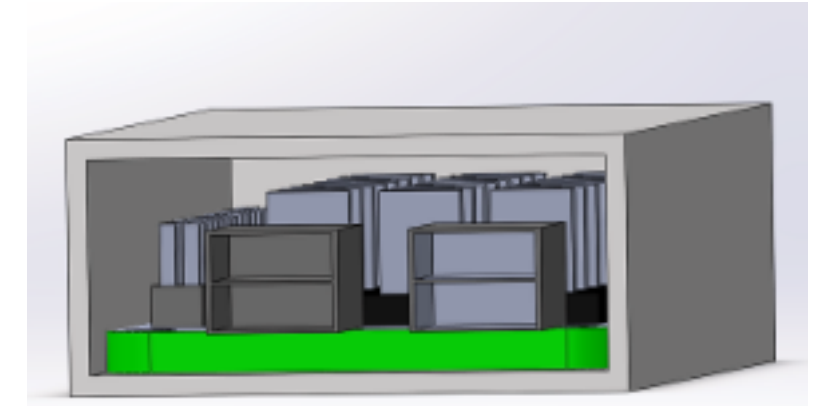
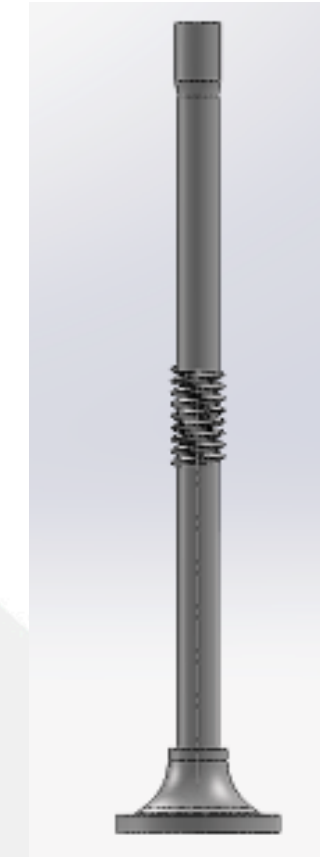
- **Antenna and Ham Radio System:** Enables robust long-distance communication, crucial in disaster response operations. 3 Material: PET plastic with copper wiring. Weight: 15 grams. Ham radio housing: PET plastic; weight approximately 730 grams.

- **I/O Ports:** Equipped with four USB interfaces via the Raspberry Pi 5.

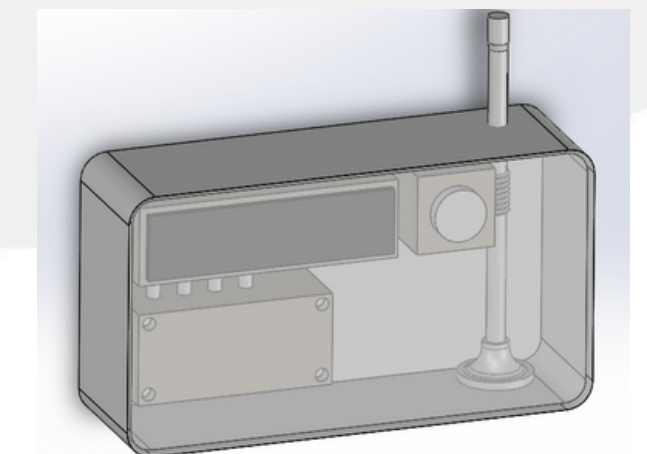
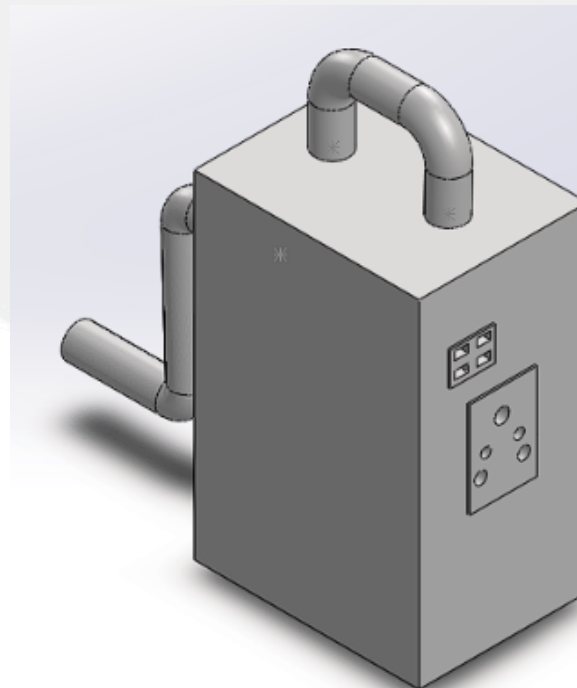
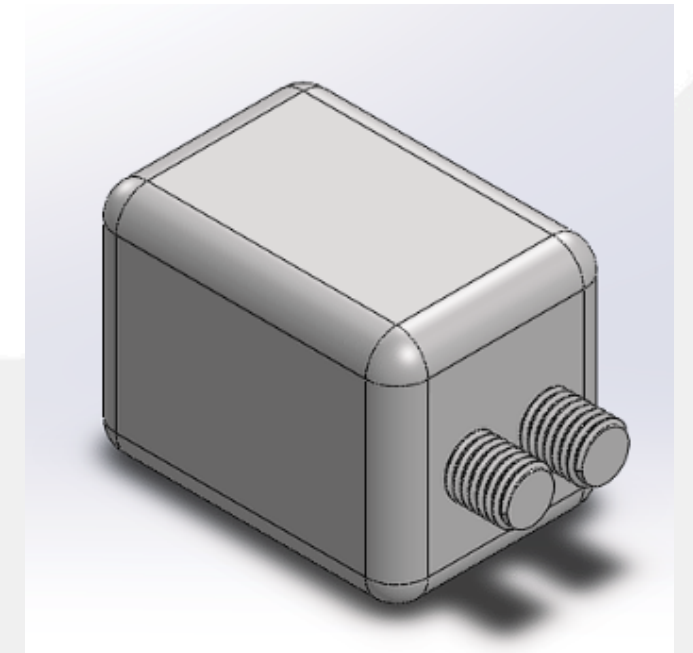
- **Manual Generator:** A hand-crank power generator ensures emergency energy availability. 5 Assembly primarily composed of PET plastic.

Weight: approximately 13.3 kg.

- **Satellite Connectivity:** Provides GPS and intermittent satellite communication for tracking and navigation.



**Raspberry-Pi
Microcontroller**



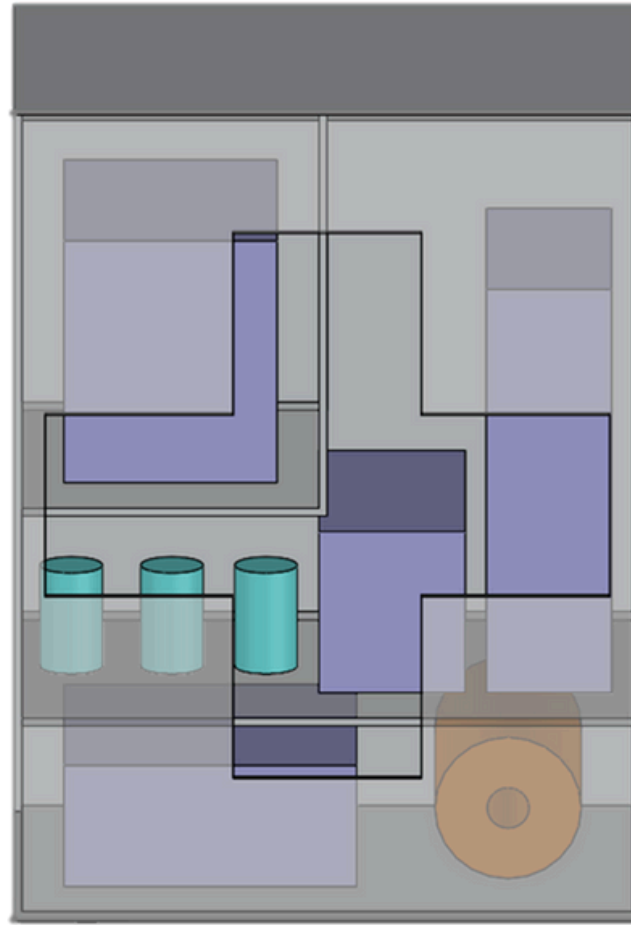
Satellite tracker



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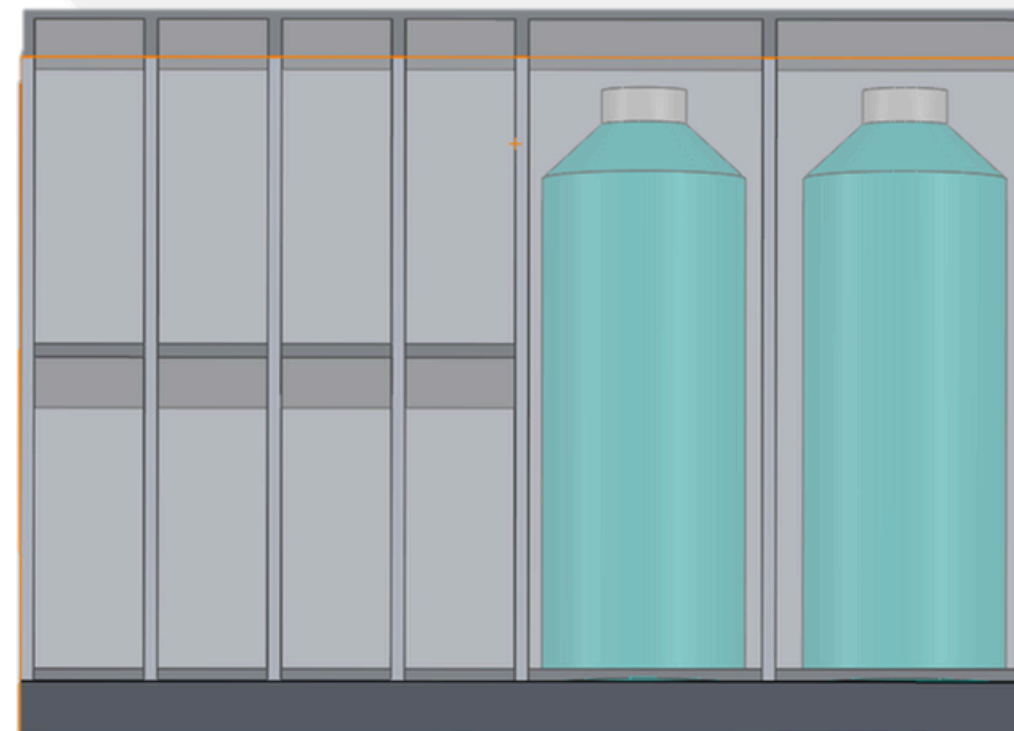
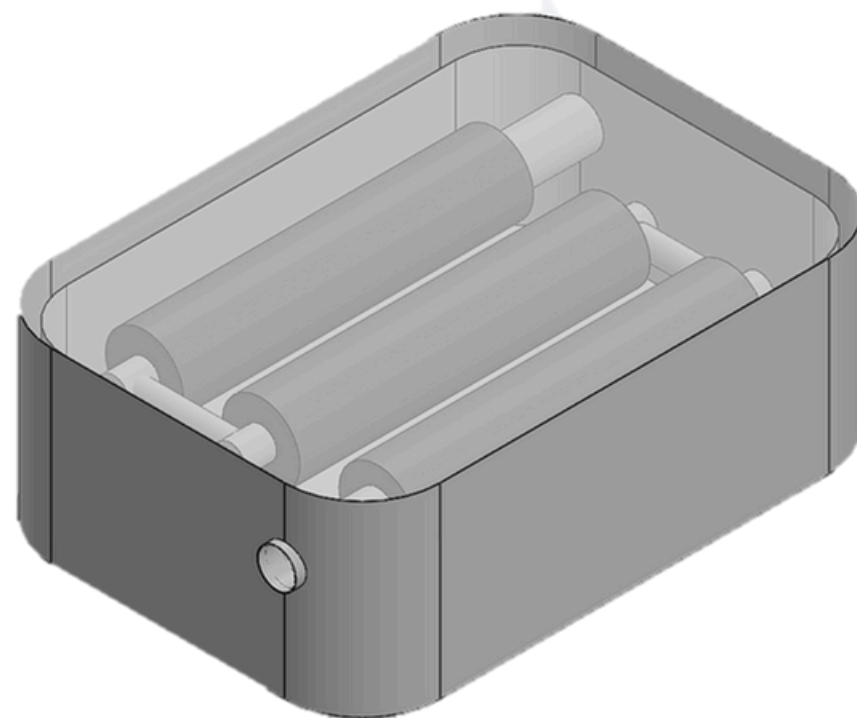
Module 3 – Medical & Sustenance

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This module addresses essential health and nutritional requirements:

- **Medical Supplies:** First-aid & trauma kits, essential meds (antibiotics, analgesics).
- **Diagnostics:** Portable BP cuff & pulse oximeter.
- **Nutrition:** Dehydrated rations & ration bars (441g comps); vitamin-fortified, long shelf-life
- **Water Purification:** 0.2µm PET microfilter (2756 g); emergency chlorination sachets (5 L yield).





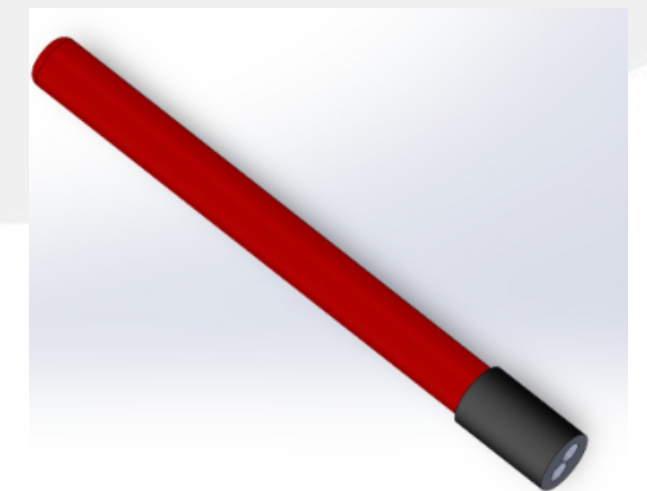
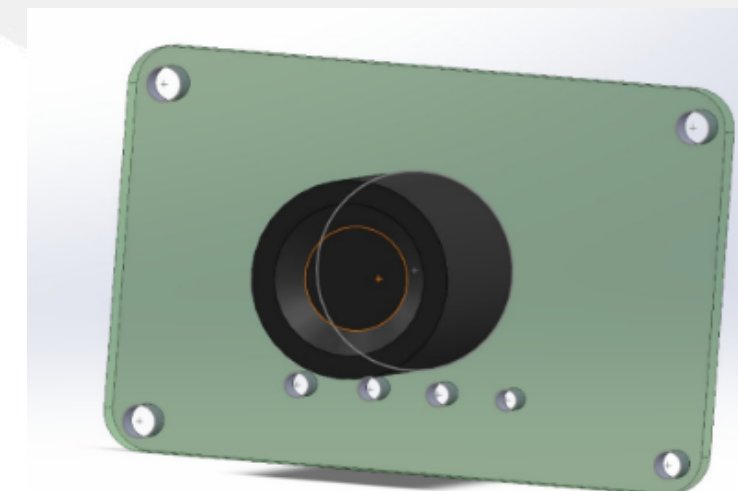
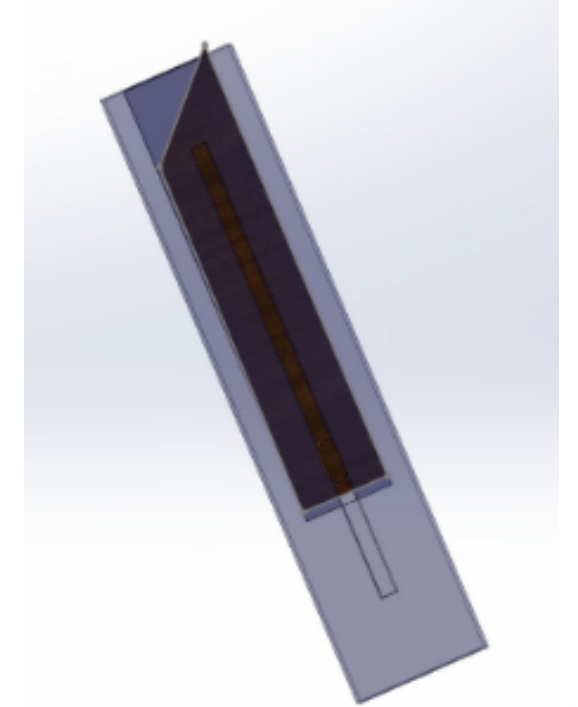
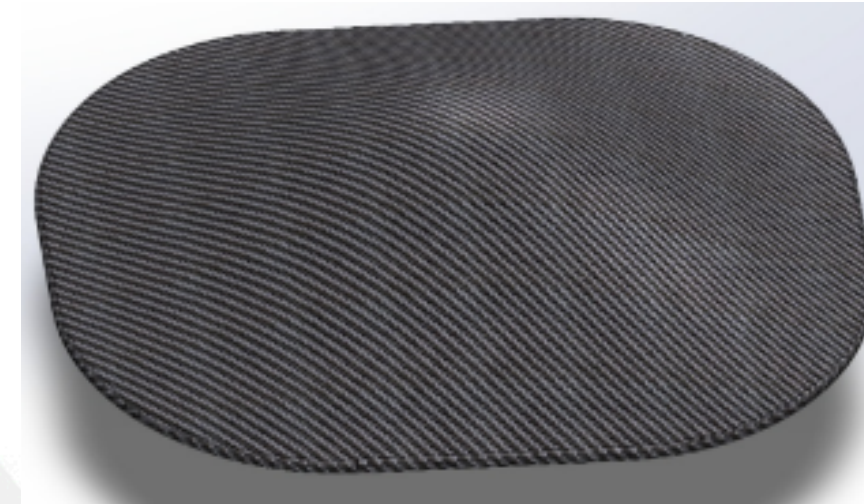
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Module 4 – Survival Kit

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This module enhances field survival and reconnaissance capabilities:

- **Essential Tools:** Utility knife, handwarmer, neoprene rope (70 g), glow stick.
- **Signal & Safety:** Flare gun (titanium alloy, 540 g), torches, life jackets, protective masks.
- **Fire & Signaling:** Ferro rod, waterproof matches, light & smoke flares.
- **Thermal Imaging:** Infrared camera (30 g) for survivor detection.





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Key Features & Design Considerations

- **Modular Architecture:** Facilitates ease of transport (62X53 cm), maintenance, and reconfiguration.
- **Hybrid Power System:** Utilizes both battery and manual energy sources for redundancy.
- **Real-Time Telemetry:** IoT dashboard monitors operational parameters in real time.
- **Robust Engineering:** Finite element simulations validate mechanical endurance under deployment stress.
- **Operating Environment:** The workstation is designed to operate in diverse disaster environments, including flood zones, earthquake-damaged areas, and regions affected by extreme weather. It functions within a temperature range of -10°C to 60°C , withstanding high humidity, potential water submersion, dust, and debris. Materials such as PET plastic and aluminum provide corrosion resistance and durability.





Calculations

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The highest velocity the body would achieve in descent is its terminal velocity,

taking 75kg, density of air as 1.225kg/m³, area of bottom surface of body as 0.217m², coefficient of drag as 0.6, the terminal velocity was found to be 30.667 m/s.

Now consider Cd as 1.75, area as 1.8237m², the terminal velocity achieved with the parachute is open is determined to be 6.2m/s.

During landing even airbags would be in action, considering a stopping distance of 0.5m, the impact force is determined.

$$v_t = \sqrt{\frac{2mg}{\rho AC_d}}$$

$$v_t = \sqrt{\frac{2 \times 75}{1.225 \times 0.217 \times 0.6}} = \sqrt{\frac{150}{0.159495}} \approx 30.667 \text{ m/s}$$

$$v_t = \sqrt{\frac{2 \times 75}{1.225 \times 1.8237 \times 1.75}} = \sqrt{\frac{150}{3.90955}} \approx 6.1941 \text{ m/s}$$

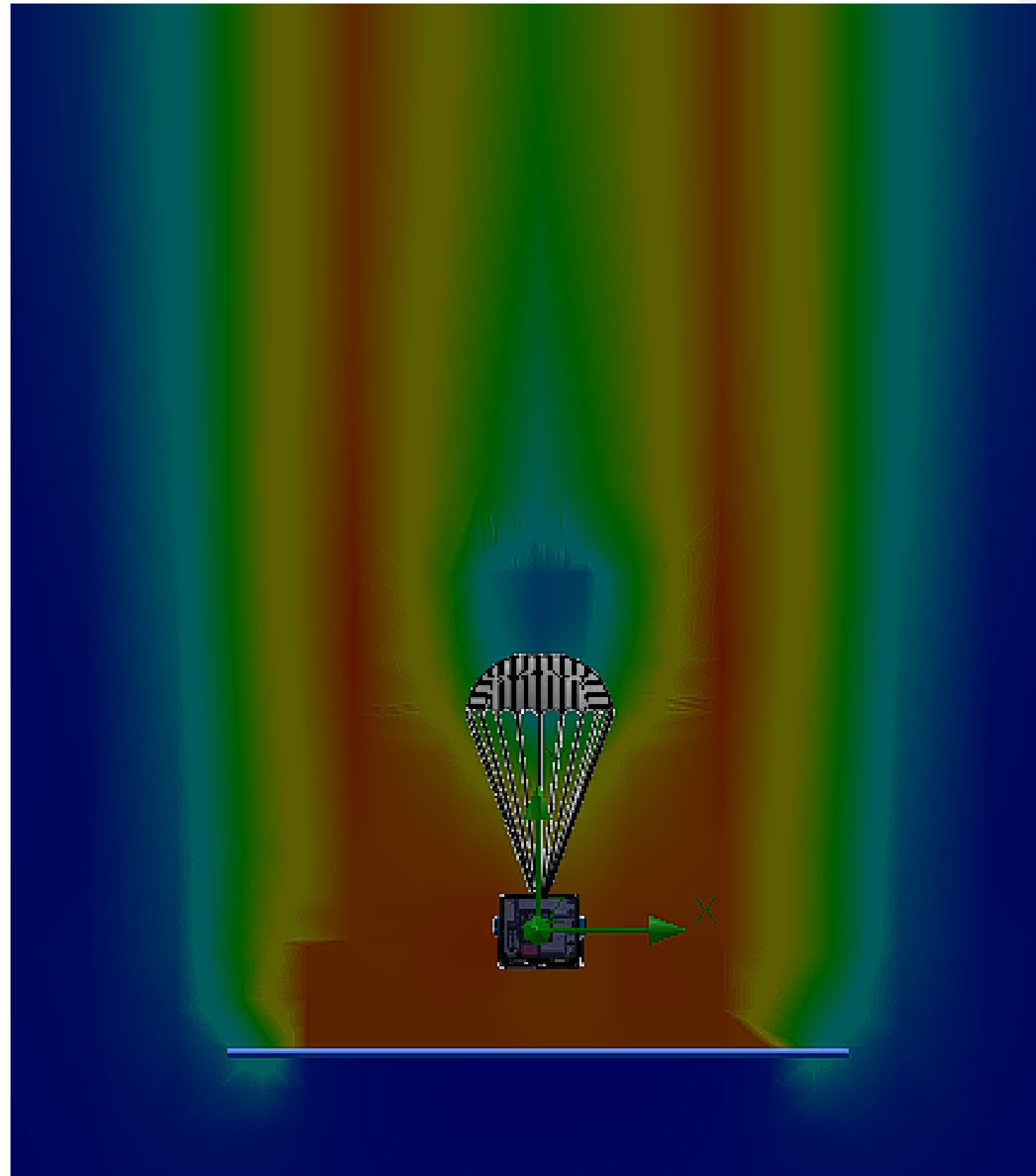
So the impact force determined is 7500N

$$a = \frac{v^2}{2d} = \frac{10^2}{2 \times 0.5} = \frac{100}{1} = 100 \text{ m/s}^2$$

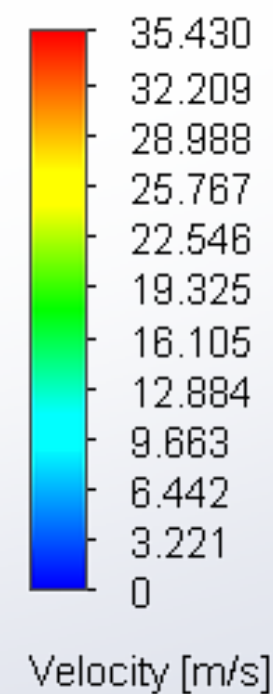
$$F = ma = 75 \times 100 = 7500 \text{ N}$$

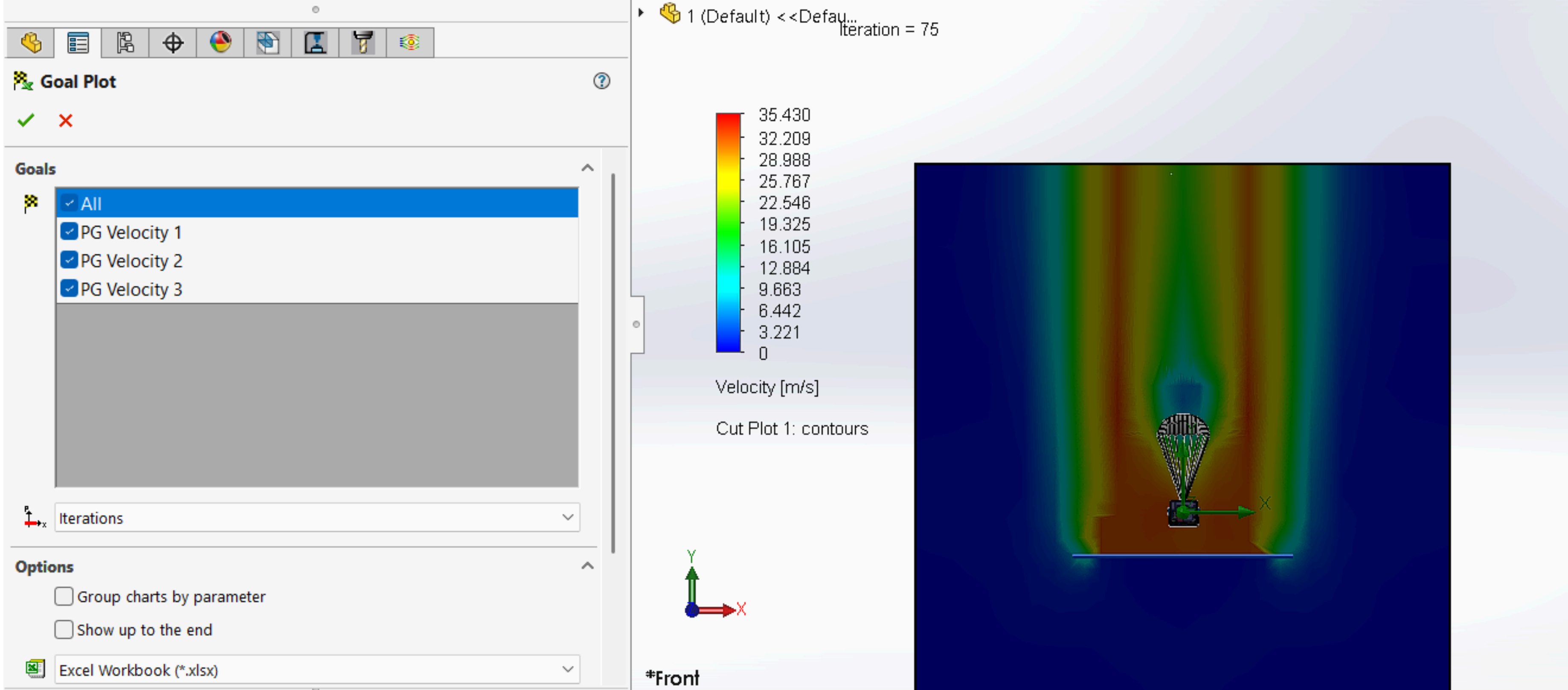


CFD Analysis of Descent Phase



- **Simulation Setup:** Terminal velocity (30.1m/s) input based on physical modeling; air density 1.225kg/m^3 ; parachute geometry (C_d 1.75, A 1.8237m^2).
- **Values:** Simulation confirms parachute system reduces v_t from 30.1m/s to $\sim 6.2\text{m/s}$, matching calculations.





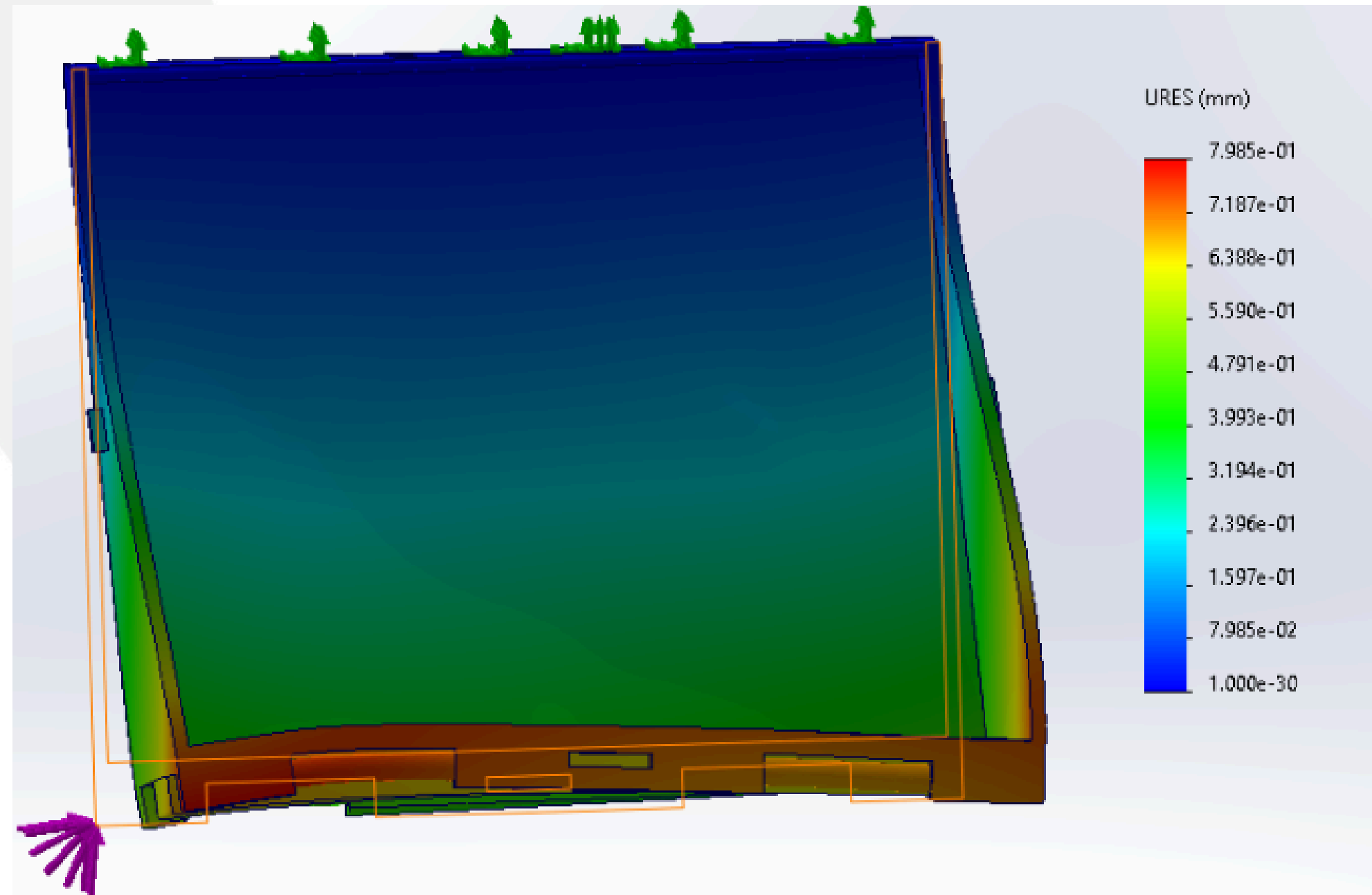
Summary

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
PG Velocity 1	[m/s]	11.391	11.615	11.352	11.800	100	Yes	0.305	1.592
PG Velocity 2	[m/s]	6.061	5.860	5.429	6.329	100	Yes	0.230	0.559
PG Velocity 3	[m/s]	5.432	5.251	4.651	5.851	100	Yes	0.412	0.413



Static Structural Analysis

- **FEA Parameters:** Al 6061 chassis T6 ($E = 69 \text{ GPa}$, $\sigma_{ny} = 276 \text{ MPa}$), mesh size 2mm, fixed base, 30 kN impact load.
- **Stress Results:** Max von Mises stress $\approx 85 \text{ MPa}$ at corner gussets (FoS = 4)
- **Validation:** FEA and calculations converge on safe impact performance, ensuring structural integrity post-drop.
- The impact is assumed to occur only at the corners, so we would be taking case of 1 corner, 2 and all corner impact ideally.
- Here is the **single corner impact**.



The max deformation is estimated to be 0.8mm, which is under the limits.

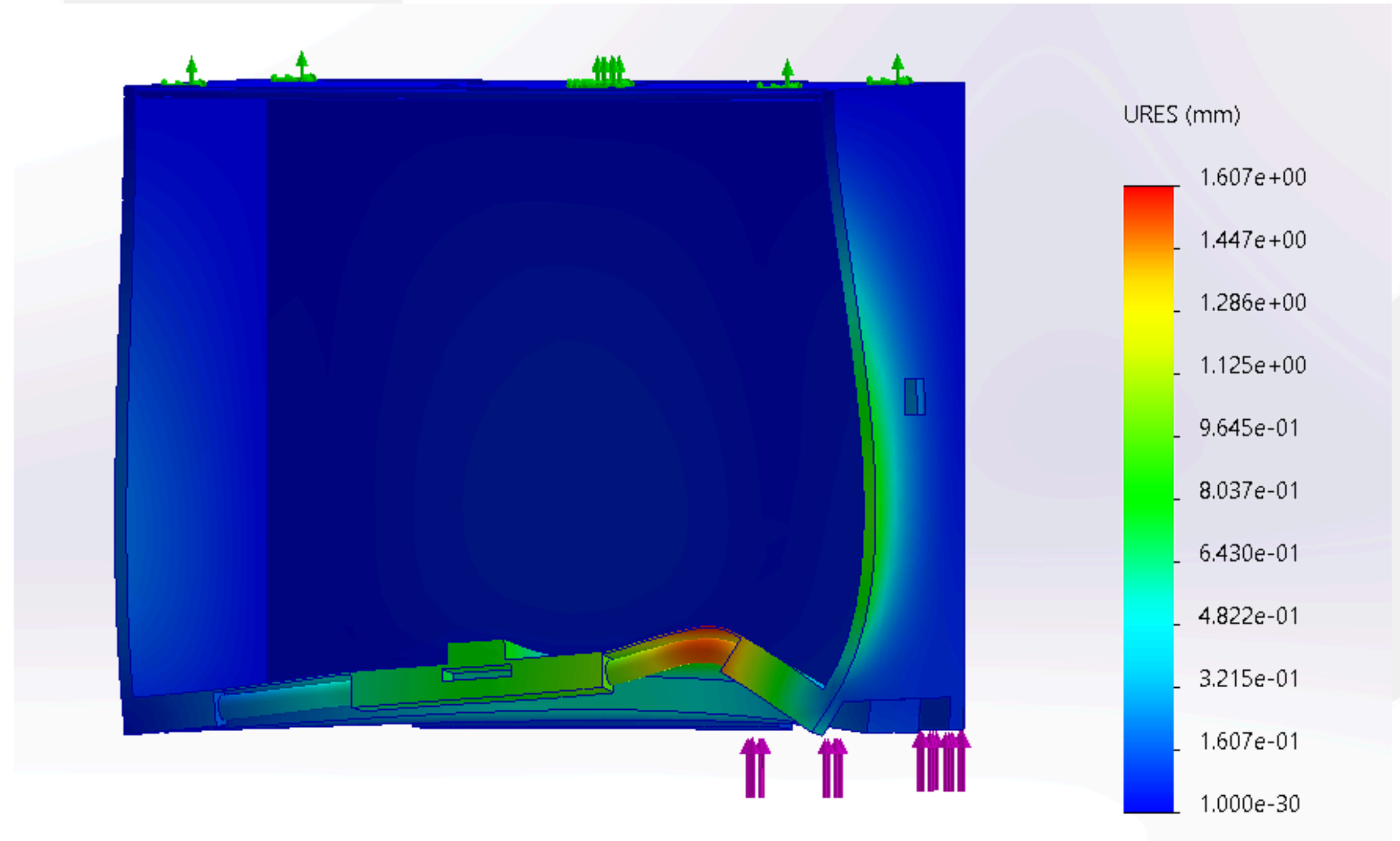


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Two corner impact:

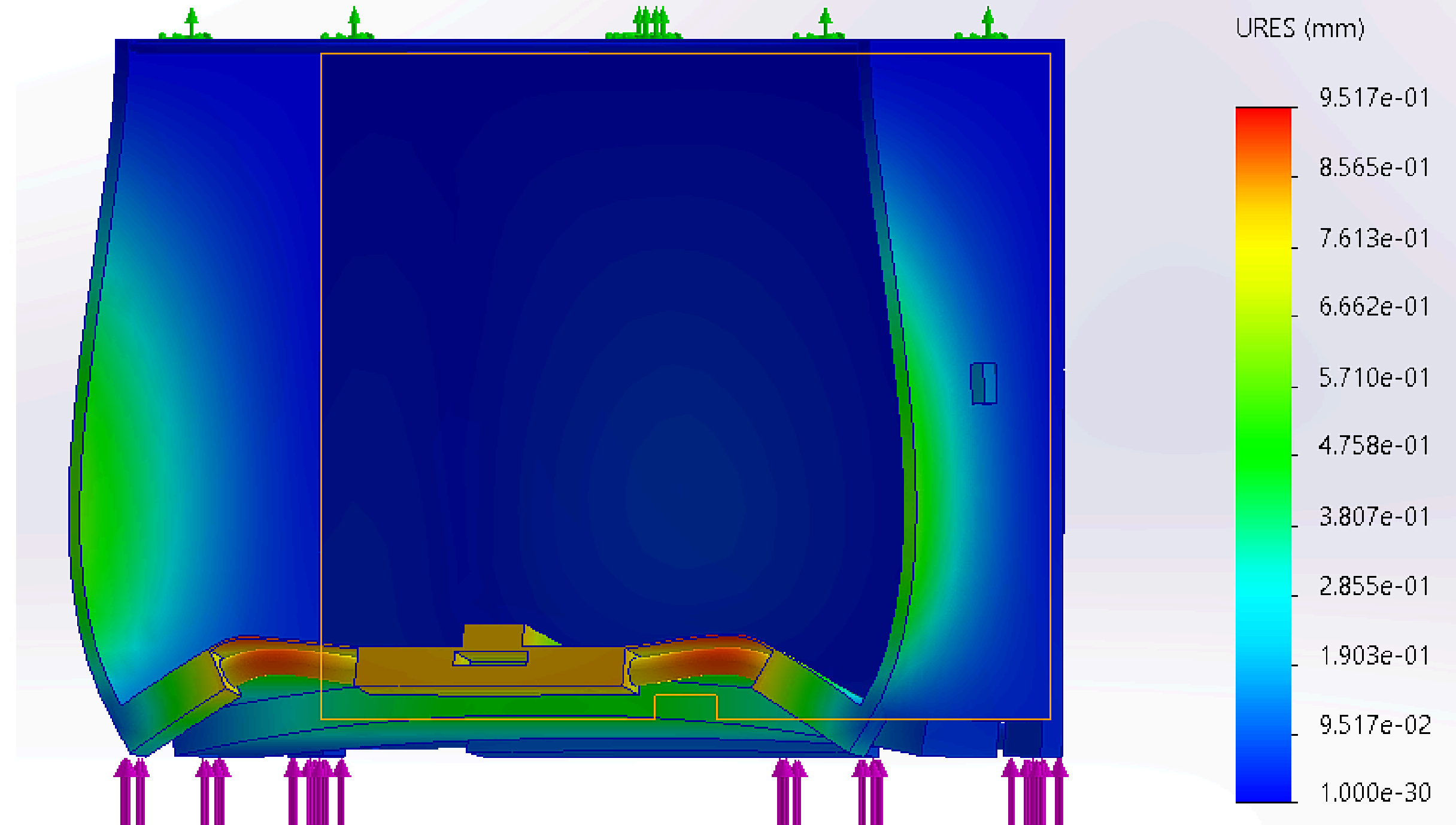
Max deformation is 1.6mm,
which is also under the limits.





All corner impact:

Max deformation is 0.9517mm,
nearly 1 mm
which is also under the limits.

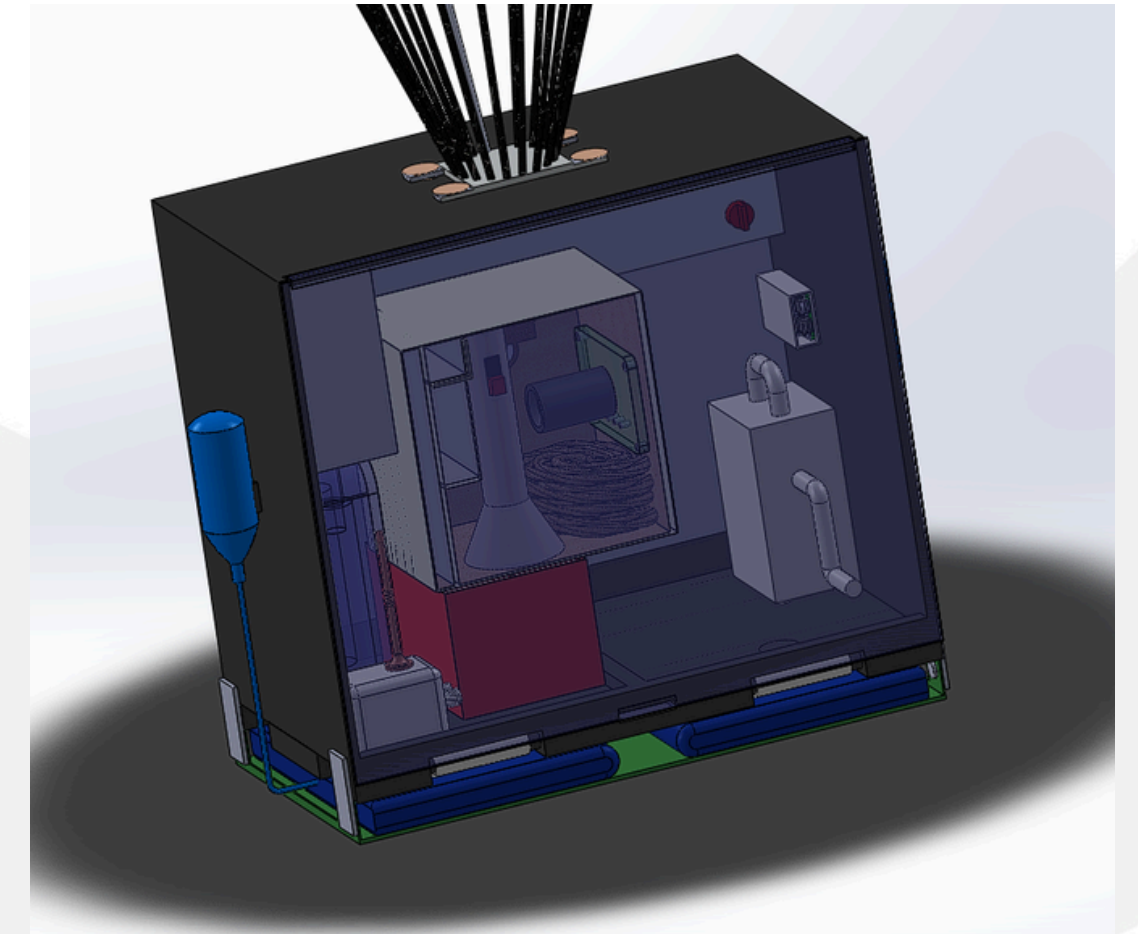
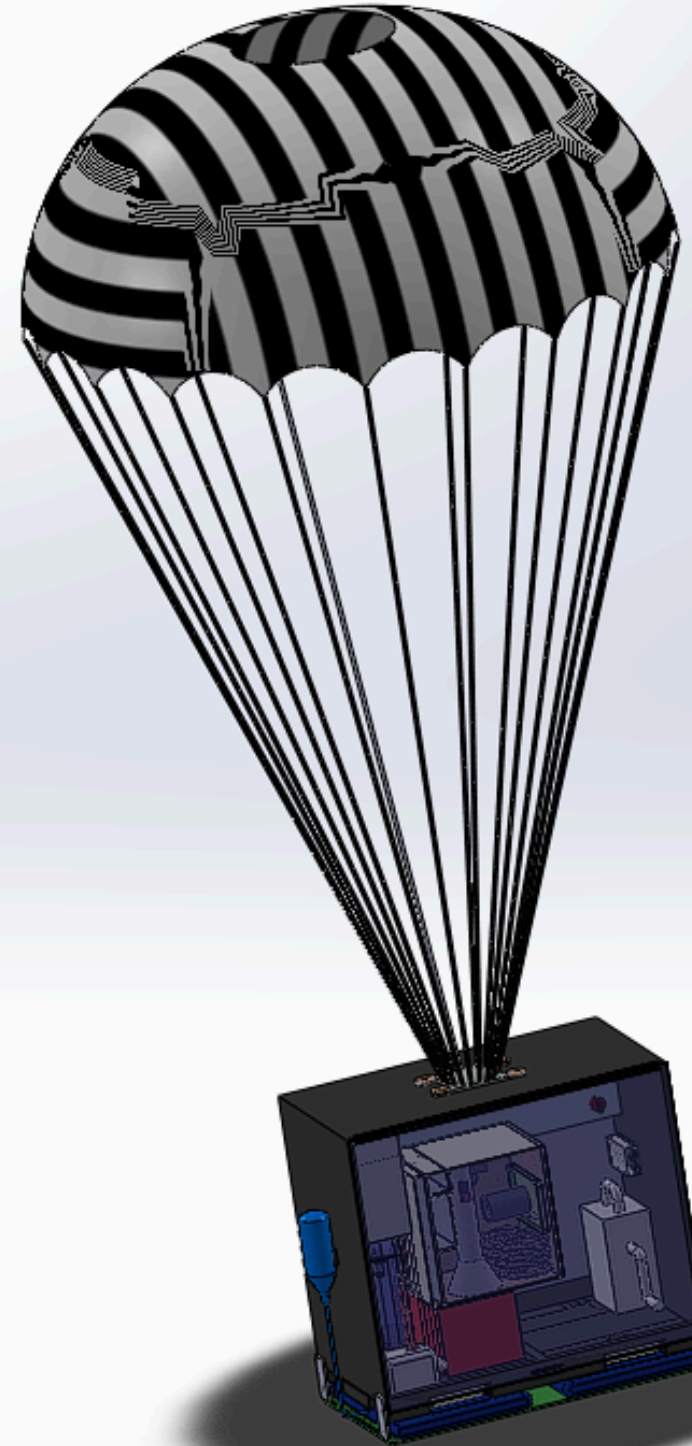




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Final assembly

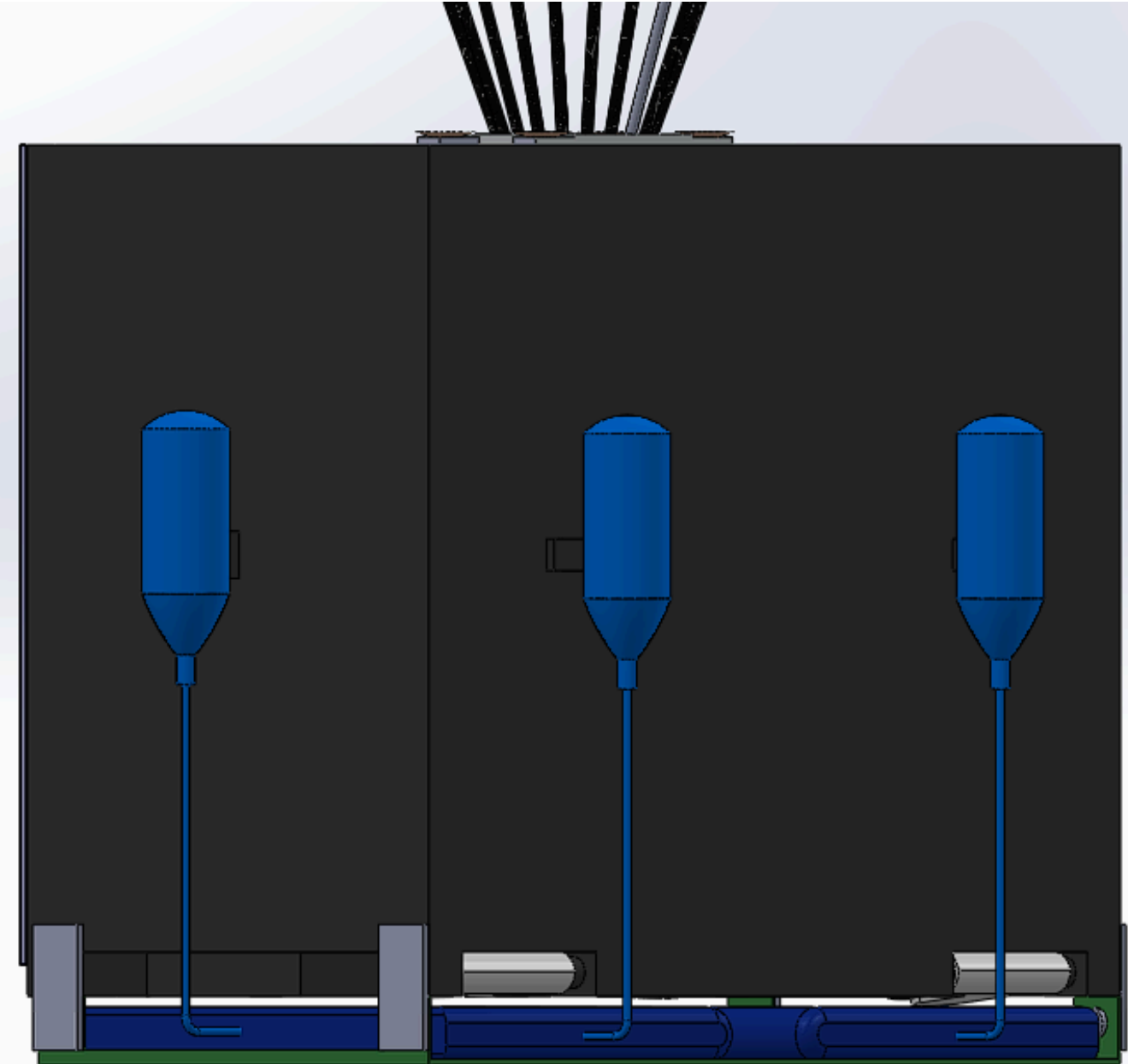
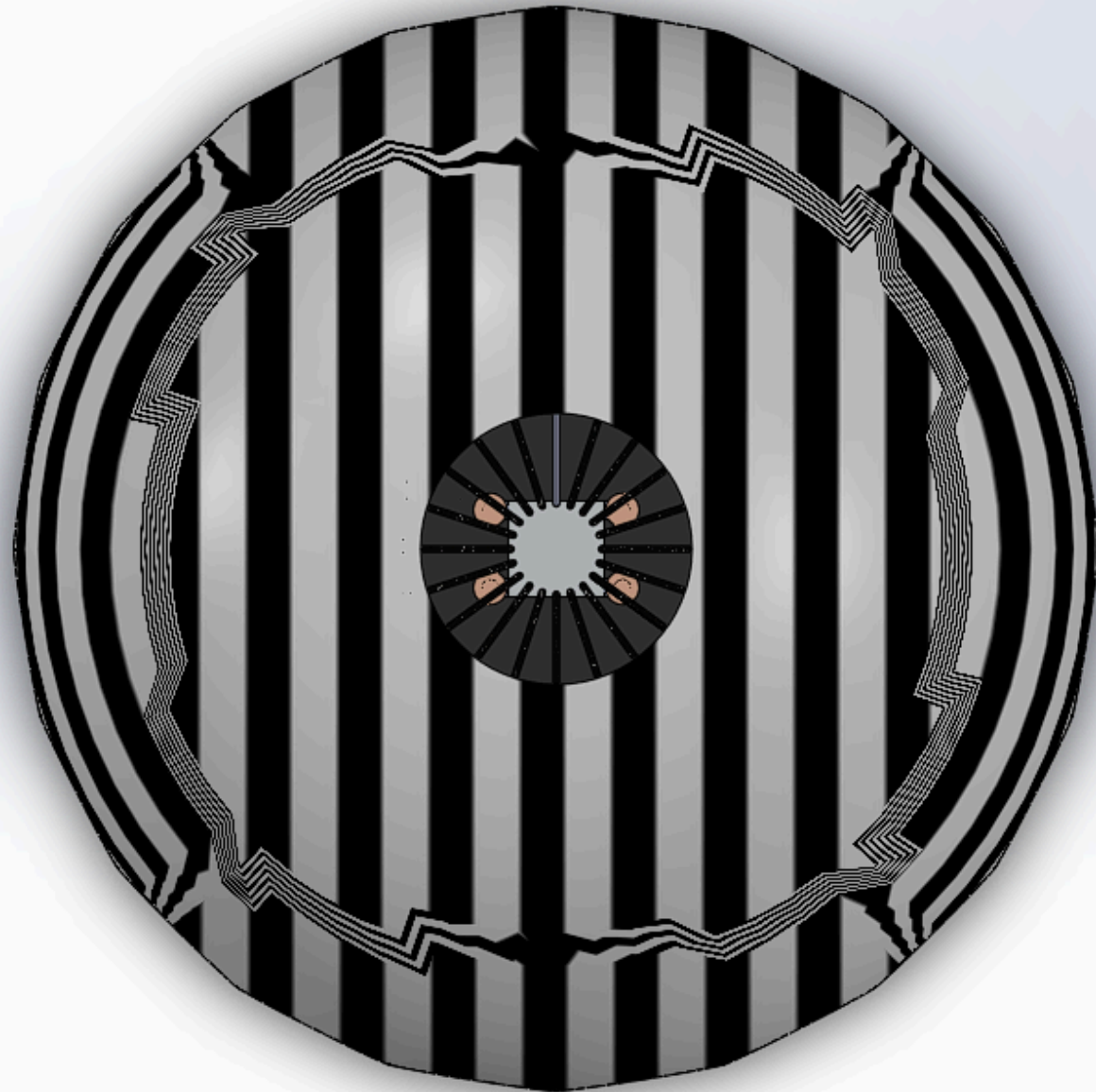
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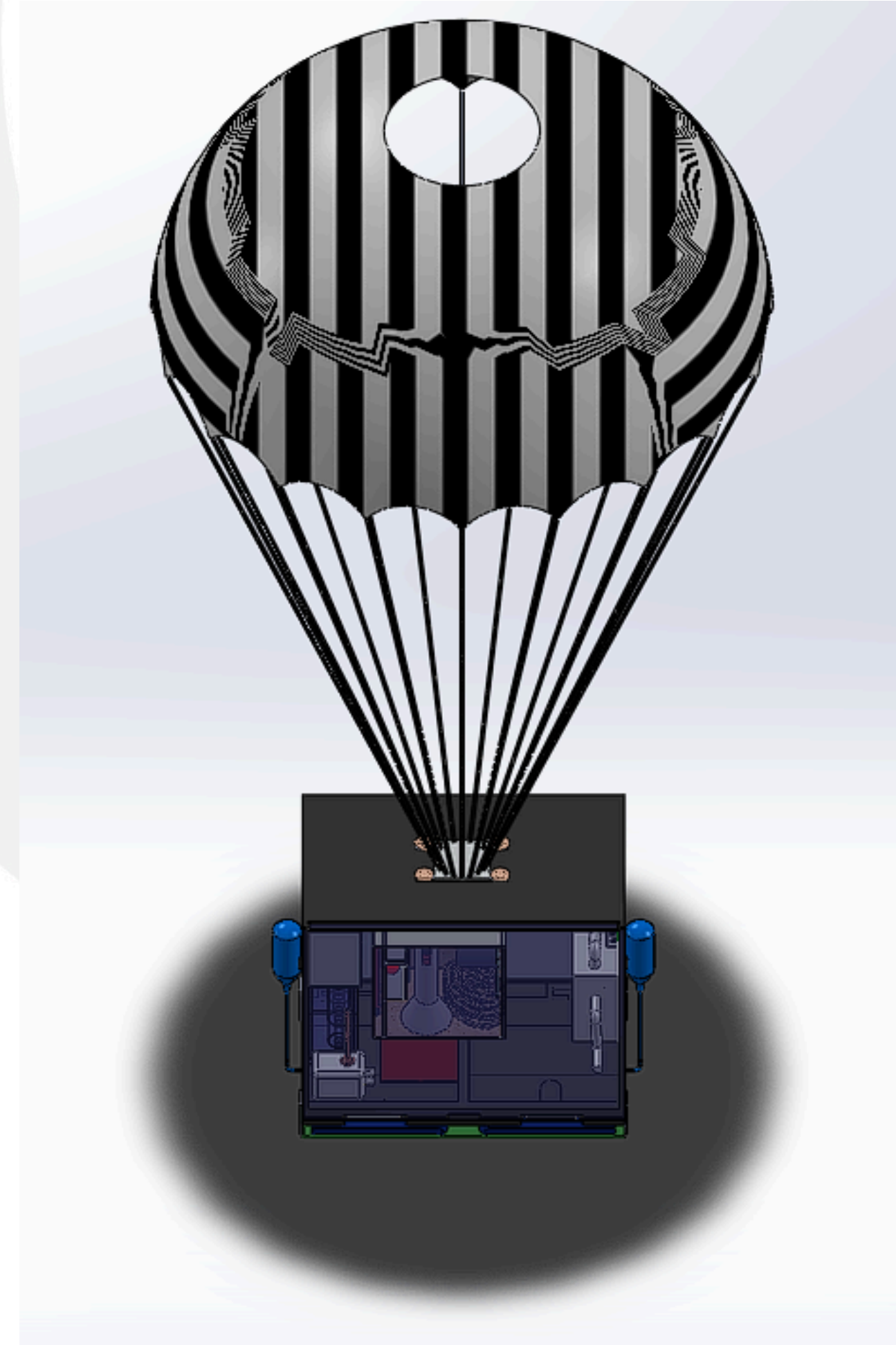
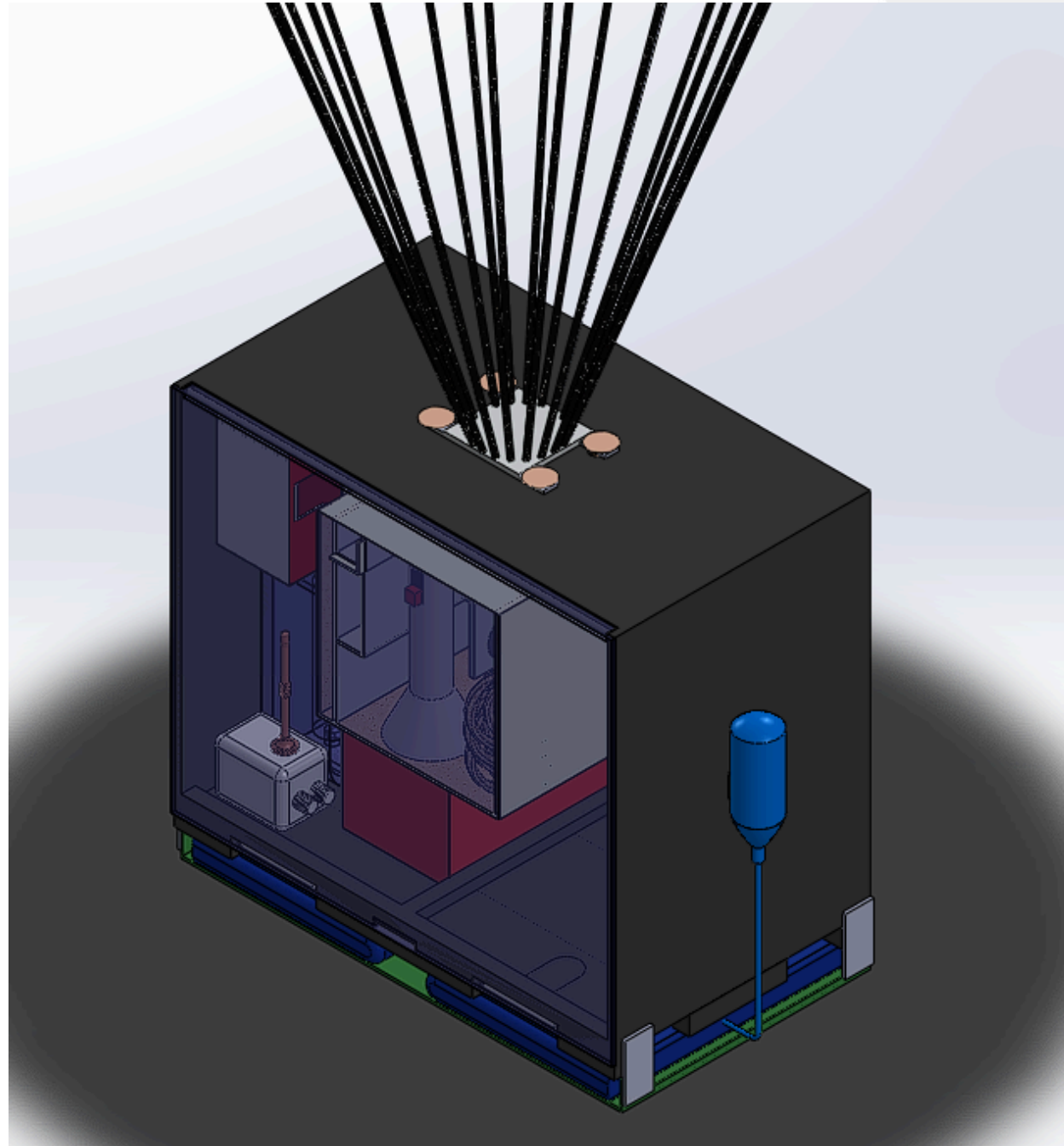
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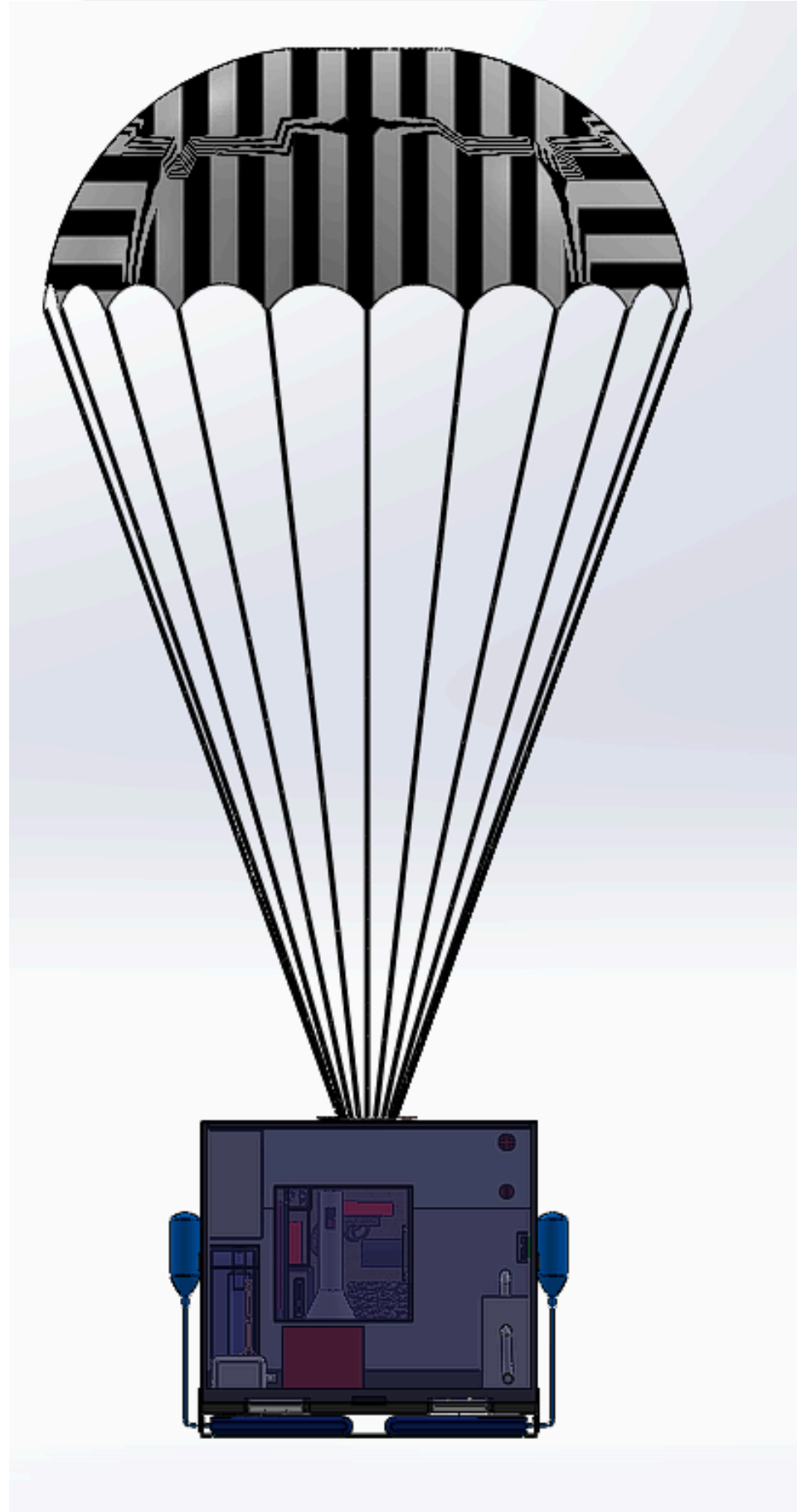
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Conclusion

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In disaster-struck regions where time and resources are scarce, the Smart Modular Workstation offers a compact, efficient, and integrated solution. By combining power, communication, medical aid, and survival tools into one air-droppable system, it ensures immediate operational readiness and supports relief teams in making informed, coordinated decisions on the ground.

Tested through simulation and grounded in practical engineering, its modular and rugged design makes it deployable in extreme environments. With the ability to adapt, power itself, and guide rescue operations from the moment it lands, the workstation exemplifies interdisciplinary innovation tailored for real-world impact.

Thank
you!